

Peidi Song

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Objective

My current research interests focus on computer system, particularly designing efficient and high-performance software service scheduling solutions for heterogeneous datacenter hardware. I have expertise in developing data structures and algorithms for networked system, including load balancing and overload control.

Education

Georgia Institute of Technology | Atlanta, GA, U.S.

Ph.D. in Computer Science

August 2024 – May 2029 (expected)

Advisors: Prof. Ahmed Saeed & Prof. Alexandros Daglis

Georgia Institute of Technology | Atlanta, GA, U.S.

Master of Science in Electrical & Computer Engineering, GPA 3.90

August 2022 – May 2024

Peking University | Beijing, China

Bachelor of Science in Computer Science, GPA 3.65

September 2018 – July 2022

Publications and Preprints

- Size Zheng, Siyuan Chen, **Peidi Song**, Renze Chen, Xiuhong Li, Shengen Yan, Dahua Lin, Jingwen Leng, and Yun Liang. "Chimera: An Analytical Optimizing Framework for Effective Compute-intensive Operators Fusion." In *2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, pp. 1113-1126. IEEE, 2023.
- **Peidi Song**, Alexandros Daglis, Michael Filler, and Ahmed Saeed. "Algorithmic Tradeoff Exploration for Component Placement and Wire Routing in Nanomodular Electronics." *arXiv preprint arXiv: 2510.03126* (2025).

Research Experiences

Resource-Aware Heterogeneous Microservice Request Load Balancing

Georgia Institute of Technology

October 2024 – Present (ongoing)

Existing load balancers treat all requests as bottlenecked on single resource, leading to underutilization when requests demand varying resources like CPU or memory. I propose a lightweight greedy heuristic to optimize resource allocation by leaving headroom for non-bottlenecked resources, thereby enhancing resource utilization. Simulations demonstrate that this approach improves utilization by up to 78% and throughput by up to 25%.

An Analytical Optimizing Framework for Effective Compute-intensive Operator Fusion

Center for Energy-efficient Computing and Applications, Peking University

October 2021 – October 2022

Existing machine learning compilers lacked both precise analysis and efficient optimization for compute-intensive operator chains on different accelerators such as CPU, GPU and NPU. I explored the optimizing framework to enhance the data locality of these operators on CPU, and achieved 2.29x speedup. *Published a conference paper in HPCA 2023.*

Algorithmic Tradeoff Exploration for Design Automation in Nanomodular Electronics

Georgia Institute of Technology

August 2023 – May 2024

I designed component placement and wire routing algorithms for nanomodular electronics, balancing tradeoffs between time-to-solution and solution quality in the manufacturing process.

Modeling and Optimization for Convolution Operator towards Ascend NPU Platform

Center for Energy-efficient Computing and Applications, Peking University

January 2022 – June 2022

I found existing Tensor operator optimization works on Ascend NPU lacked a performance model that helped developers to reason about expected performance. I presented an abstraction of the hardware architecture and an accurate performance cost model for convolution operators on the platform, and achieved 4.76x speedup.

Skills

Programming: C, C++, Python, Cuda, MPI, OpenMP, Verilog, HLS, Pascal

Software: DPDK kernel-bypass network stack, Apache Thrift RPC framework, Docker and Kubernetes container management tool for cloud applications